Yanning DAI

PhD Student, King Abdullah University of Science and Technology (KAUST)

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Education

Ph.D. in Computer Science *Jun 2024 – Present*
King Abdullah University of Science and Technology (KAUST), Saudi Arabia GPA: 3.9/4.0
Advisor: Prof. Jürgen Schmidhuber

M.S in Electrical Engineering *Sep 2020 - Jul 2023*
Beihang University, China GPA: 3.8/4.0
Advisor: Prof. Shuo Gao

B.E. in Electrical Engineering *Sep 2016 - Jul 2020*
Beihang University, China GPA: 3.8/4.0

Research Interests

Self-improving Agents, Reinforcement Learning, Robotic Learning

Selected Publications

1. *Efficient Morphology–Control Co-Design via Stackelberg Proximal Policy Optimization.*
Yanning Dai*, Yuhui Wang*, Dylan R. Ashley, and Jürgen Schmidhuber. **ICLR**, 2026.

2. *OASIS: Online Anchor-calibrated Self-Improvement System.*
Wenyi Wang, Firas Laakom, Li Nanbo, **Yanning Dai**, Yuhui Wang, Yifan Xu, Dmitrii Khizbullin, Piotr Piękos, Yilan Zhang, Yimeng Chen, Alfonso Amayuelas, Mingchen Zhuge, and Jürgen Schmidhuber. **EMNLP Findings**, 2026.

3. *A Unified Framework for Rethinking Policy Divergence Measures in GRPO.*
Qingyuan Wu, Yuhui Wang, Simon Sinong Zhan, **Yanning Dai**, Shilong Deng, Sarra Habchi, Qi Zhu, Matthias Gallé, and Chao Huang. arXiv:2602.05494, 2026.

4. *Wearable-Informed Generative Digital Avatars Predict Task-Conditioned Post-Stroke Locomotion.*
Yanning Dai, Chenyu Tang, Ruizhi Zhang, Wenyu Yang, Yilan Zhang, Yuhui Wang, Junliang Chen, Xuhang Chen, Ruimou Xie, Yangyue Cao, and Shuo Gao. arXiv:2512.14329, 2026.

5. *Towards an Extremely Robust Baby Robot With Rich Interaction Ability for Advanced Machine Learning Algorithms.*
Mohannad Alhakami*, Dylan R. Ashley*, Joel Dunham*, **Yanning Dai**, Francesco Faccio, Eric Feron, and Jürgen Schmidhuber. **IROS**, 2025.

6. *Tactile and Vision Perception for Intelligent Humanoids.*
Shuo Gao, **Yanning Dai**, and Arokia Nathan. **Advanced Intelligent Systems**, 2022. **[Cover Article]**.

7. *Advanced Electronics and Artificial Intelligence: Must-have Technologies toward Human Body Digital Twins.*
Yanning Dai, Jiaqi Wang, and Shuo Gao. **Advanced Intelligent Systems**, 2022.

8. *Wearable Systems based Gait Monitoring and Analysis.*
Shuo Gao, Junliang Chen, **Yanning Dai**, and Boyi Hu. **Springer Nature**, 2022. **[Book]**.

9. *A Flexible Multi-Functional Smart Skin for Force, Touch Position, Proximity, and Humidity Sensing for Humanoid Robots.*
   Yanning Dai and Shuo Gao. **IEEE Sensors Journal**, 2021.

10. *Force Touch and Machine Learning Based Smart Sensing Techniques for Interactive Displays.*
    Shuo Gao, **Yanning Dai**, Junliang Chen, Anbiao Huang, and Yong Liu. **SID Symposium Digest of Technical Papers**, 2021.

11. *A PVDF/Au/PEN Multifunctional Flexible Human–Machine Interface for Multidimensional Sensing and Energy Harvesting for the Internet of Things.*
    Yanning Dai, Junliang Chen, Wenbin Tian, Lijun Xu, and Shuo Gao. **IEEE Sensors Journal**, 2020.

12. *Multi-Functional Smart Skin for Multi-Dimensional Perception for Humanoid Robots.*
    Yanning Dai and Shuo Gao. **IEEE FLEPS**, 2020.

13. *A PVDF/Au/PEN Multifunctional Flexible Human–Machine Interface for 2D Position, Force, Hover, Surface Curvature Sensing and Energy Harvesting.*
    Yanning Dai and Shuo Gao. **IEEE FLEPS**, 2019.

*Equal contribution.

Experience

Research Intern *Jan 2024– May 2024*
Beijing Tsinghua Changung Hospital, Tsinghua University

Built data-driven world models for patient-specific recovery dynamics on large-scale clinical datasets.

Research Assistant *Jun 2023– Jan 2024*
Advanced Sensing and Intelligent Systems Lab, Beihang University

Developed reinforcement learning and generative models for personalized digital human systems and behavior generation in stroke rehabilitation.

Research Intern *Jun 2022– Jun 2023*
National Engineering Research Center for Ophthalmic Diagnosis

Modeled patient-specific temporal dynamics of myopia progression from longitudinal ophthalmic data.

Selected Honors

1. Rising Stars in AI, KASUT AI Initiative (30 Awardees Worldwide) *Apr 2024*

2. National Scholarship, Ministry of Education of China (Top 5%) *Dec 2022*

3. The 3rd Class Award of Science and Technology, China Instrument and Meter Society (National S&T Award, Rank: 2/6) *Oct 2022*

4. China Instrument and Control Society Scholarship, 1st Class Award, China Instrument and Meter Society (Top Society Award, Top 1%) *Nov 2019*

Skills

Programming: Python, C++, C, C#, HTML, Shell, Matlab

Project / Lib Experiences: PyTorch, TensorFlow, Transformers, Distributed Training (FSDP), Accelerated Inference (SGLang, vLLM), ROS, Robot Simulation (MuJoCo, Gazebo, Unity), Unitree SDK

Languages: English (Fluent), Mandarin (Native)

Teaching

School of Instrumentation and Optoelectronic Engineering, Beihang University:

Teaching Assistant, IntelliSense and Autonomous Systems *Spring 2023*
Teaching Assistant, Introduction to Smart Sensing Systems *Fall 2022*

Services

Served as a reviewer for NeurIPS, ICLR, EMNLP, and IEEE FLEPS, as well as for journals including Expert Systems with Applications, IEEE Sensors Journal, IEEE Sensors Letters, and Frontiers in Electronics.